

A couple power and free horizontal roller turns have free track channel gap (toe-to-toe) spread beyond the designed channel gap of 1-3/4". Repairs can be made as described under the **Straight Track Assemblies** (See paragraph IVC).

F. Horizontal Traction Wheel Turn Assemblies (Power and Free/Power Only):

Most power only horizontal traction wheel turns are in satisfactory condition. The power track (4" I-beam) has minimal wear. Also, the traction wheels that were observed had minimal rim wear. Random wear checks were made to the power track portions of accessible horizontal traction wheel turns on all conveyors.

The majority of power only traction wheel turns appeared to have adequate hub bearing lubrication at this time. Hub bearings should be lubricated (greased) at least annually, and quite possibly semiannually. Under normal operating and environmental conditions two to three pumps of a hand-held grease gun or approximately one to two ounces of grease should be sufficient. Attention to over greasing should be observed, as excess grease will blow out the bearing seals and may damage the seals.

G. Drive Assemblies (Linear):

All conveyors use linear type drives which appeared to be in satisfactory condition, operating properly and running somewhat smoothly. However, a few drives are in need of component replacement and/or repair. Some cat chains have a build-up of grease and powder on the roller chain side links and rollers. This build-up can prohibit flow of oil to the proper wear points. Several drive and take-up sprockets appeared to have moderate to heavy wear. These sprockets should be replaced and realigned with cat and conveyor chain.

Plant maintenance should routinely inspect all linear floating drives, including pivots, back-up bars, roller nests and overload limit switches should be checked for proper operation. Cat chains, drive and take-up sprockets and back-up bars should be checked for wear and alignment in relationship to the conveyor chain. As mentioned under **Conveyor Chain**, caterpillar drive cat chains should be replaced and lower units adjusted and/or rebuilt when new conveyor chain is installed, **as worn cat chain will not mesh properly with new conveyor chain.**

All drive units were equipped with a limit switch torque overload protection system. The current drive spring preload and torque overload cutout settings (in pounds of chain pull) have been calculated based on field measurements and are listed in **Section V**, Detailed Results and Recommendations. Please note that these settings are for reference purposes only and are **approximate**, as dimensional data was obtained with a tape measure during the inspection.

Drives which are floating substantially can have additional drive spring preload added in order to keep the floating frame from bouncing and causing nuisance torque overload cutouts. Adding additional drive spring preload will not affect the torque overload limit switch setting or cutout point; however, changing compression spring rates will. Drive units requiring the drive spring to be replaced with a higher compression rate spring in order to keep them from floating, must be recalibrated. The torque overload limit switch setting should not be altered once a drive has been calibrated; even if additional drive spring preload is added. Plant Maintenance should routinely check limit switch operation as potential reducer damage and/or serious drive failures could occur if it does not operate properly in a torque overload situation.

Some cat chains have a moderate to heavy buildup of grease. **Cat dogs** can be greased to assist meshing with the conveyor chain; however, **cat chains should not be greased**. Grease or other debris on the cat chain will prevent oil from penetrating the chain rollers. As mentioned in the **Conveyor Chain** section, caterpillar drive cat chains should be replaced and lower units adjusted and/or rebuilt when new conveyor chain is installed, as **worn cat chains will not mesh properly with new conveyor chain**. Prior to installing a new cat chain, it should be soaked in heavy gear oil (in most cases new cat chain ships with only a rust preventive). Webb cat chains are soaked in MWO 30 oil for approximately 8 to 10 hours (Webb standard). Soaking the cat chain will allow oil to penetrate the roller bearings, which cannot be reached from external oiling. Once in operation, cat chains should be oiled and cat dogs greased (optionally) on a regular basis.

A few drives had worn cat chain, drive and take-up sprockets. An improperly positioned backup bar can cause wear and/or damage to the conveyor chain, caterpillar chain, drive or take-up sprockets, and power track. Periodic adjustments are recommended. Worn cat chains, backup bars and sprockets should be replaced. While finding the ideal setup and alignment, including the amount of cat dog engagement to the conveyor chain for smooth operation may be somewhat difficult at best, a set of backup bar adjustment guidelines is listed on the next page. See **Figure 5, 4" Overhead P&F Caterpillar Drive**, for reference.

Backup Bar Adjustment Guidelines

1. Follow all safety procedures to ensure the conveyor is adequately de-energized and locked out.
2. Restrain the conveyor chain from running away by means of a high strength coil chain with a minimum working load of 7500 lb. **This is critical due to the possibility of imbalanced vertical curve loads which would back drive the reducer when the motor brake is released, causing a chain run away condition.**
3. Position one caterpillar chain dog as shown in **Figure 4**. Do this by manually releasing the brake and turning the reducer input shaft sheave.
4. Slacken the caterpillar chain slightly by loosening the take-up adjustment screw.
5. Position the backup bar using the adjustment screws at the lead end and exit end to maintain a 1/8" clearance between the top of the caterpillar dogs and the face of the backup roller. This clearance should be checked with all caterpillar dogs.
6. Tighten caterpillar chain with take-up adjustment screw to reduce the horizontal deflection of the caterpillar chain to not less than 1/4" and not more than 3/8". Recheck the cat dog position.
7. Check the alignment of the backup bar and sprockets. Proper alignment will allow the sprocket teeth to engage the caterpillar chain without "polishing" the top or bottom surface of the sprocket teeth.
8. Run the conveyor. Listen and check for a "Slapping Action and/or Sound" which may occur from the caterpillar chain dog wiping away from the conveyor chain. If this is evident, move the exit end of the backup bar away from the conveyor chain approximately 1/8".
9. Retighten the caterpillar chain by using the take-up adjustment screw.
10. Proper tension should also be checked when the conveyor is running by observing the caterpillar dog when it is engaged with the conveyor chain. The edge of the caterpillar chain side bar should be parallel with the backup bar. If it is not, recheck the caterpillar chain tension by repeating steps 1 - 9 above.

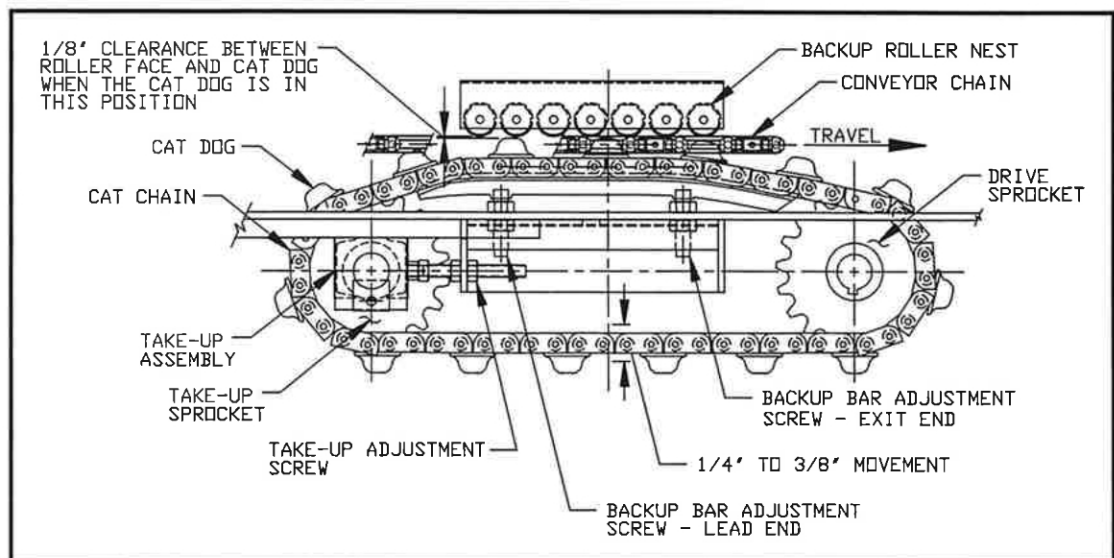


Figure 5. 4' Overhead P&F Caterpillar Drive